Assignement 1

1. Define a schema for a Library Management System with the following entities:
   - Books
   - Authors
   - Members
   - Borrow_Records

```
mysql> create table Authors(author_id primary key int, name varchar(100),country varcha
);
```

```
mysql> desc Authors;
+-----------+--------------+------+-----+---------+-------+
| Field     | Type         | Null | Key | Default | Extra |
+-----------+--------------+------+-----+---------+-------+
| author_id | int          | NO   | PRI | NULL    |       |
| name      | varchar(100) | YES  |     | NULL    |       |
| country   | varchar(50)  | YES  |     | NULL    |       |
+-----------+--------------+------+-----+---------+-------+
3 rows in set (0.08 sec)
```

```
mysql> create table Books(book_id int primary key,author_id int ,title varchar(150),for
key (author_id) references  Authors(author_id),published_year year,available_copies int
Query OK, 0 rows affected (0.17 sec)

mysql> desc Books;
+------------------+--------------+------+-----+---------+-------+
| Field            | Type         | Null | Key | Default | Extra |
+------------------+--------------+------+-----+---------+-------+
| book_id          | int          | NO   | PRI | NULL    |       |
| author_id        | int          | YES  | MUL | NULL    |       |
| title            | varchar(150) | YES  |     | NULL    |       |
| published_year   | year         | YES  |     | NULL    |       |
| available_copies | int          | YES  |     | NULL    |       |
+------------------+--------------+------+-----+---------+-------+
5 rows in set (0.01 sec)
```

```
mysql> CREATE TABLE Members(member_id int PRIMARY KEY,name varchar(100),email VARCHAR(1
hone VARCHAR(15));
Query OK, 0 rows affected (0.05 sec)

mysql> desc Members;
+-----------+--------------+------+-----+---------+-------+
| Field     | Type         | Null | Key | Default | Extra |
+-----------+--------------+------+-----+---------+-------+
| member_id | int          | NO   | PRI | NULL    |       |
| name      | varchar(100) | YES  |     | NULL    |       |
| email     | varchar(100) | YES  |     | NULL    |       |
| phone     | varchar(15)  | YES  |     | NULL    |       |
+-----------+--------------+------+-----+---------+-------+
4 rows in set (0.02 sec)
```

```
mysql> CREATE TABLE Borrow_Records(record_id INT PRIMARY KEY,member_id int, book_id int
row_date DATE,return_date DATE,FOREIGN KEY(member_id) REFERENCES Members(member_id),FOR
KEY(book_id) REFERENCES Books(book_id));
Query OK, 0 rows affected (0.09 sec)

mysql> desc Borrow_Records;
+-------------+------+------+-----+---------+-------+
| Field       | Type | Null | Key | Default | Extra |
+-------------+------+------+-----+---------+-------+
| record_id   | int  | NO   | PRI | NULL    |       |
| member_id   | int  | YES  | MUL | NULL    |       |
| book_id     | int  | YES  | MUL | NULL    |       |
| borrow_date | date | YES  |     | NULL    |       |
| return_date | date | YES  |     | NULL    |       |
+-------------+------+------+-----+---------+-------+
5 rows in set (0.01 sec)
```

6. Modify the Books table to add a column genre of type VARCHAR(50).

```
mysql>  ALTER TABLE Books ADD genre VARCHAR(50);
Query OK, 0 rows affected (0.23 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc Books;
+------------------+--------------+------+-----+---------+-------+
| Field            | Type         | Null | Key | Default | Extra |
+------------------+--------------+------+-----+---------+-------+
| book_id          | int          | NO   | PRI | NULL    |       |
| author_id        | int          | YES  | MUL | NULL    |       |
| title            | varchar(150) | YES  |     | NULL    |       |
| published_year   | year         | YES  |     | NULL    |       |
| available_copies | int          | YES  |     | NULL    |       |
| genre            | varchar(50)  | YES  |     | NULL    |       |
+------------------+--------------+------+-----+---------+-------+
6 rows in set (0.04 sec)
```

7. Write the SQL command to drop the Borrow_Records table.

```
mysql> drop table Borrow_Records;
Query OK, 0 rows affected (0.02 sec)

mysql> desc Borrow_Records;
ERROR 1146 (42S02): Table 'library.borrow_records' doesn't exist
```

8. Insert 3 records into the Authors table.

```
mysql> INSERT into Authors VALUES(101,'tejas','India'),(102,'khushi','India'),(103,'roh
canada');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> select * from Authors;
+-----------+--------+---------+
| author_id | name   | country |
+-----------+--------+---------+
|       101 | tejas  | India   |
|       102 | khushi | India   |
|       103 | rohan  | canada  |
+-----------+--------+---------+
3 rows in set (0.00 sec)
```

9. Insert 5 books into the Books table.

```
mysql> INSERT into Books(book_id,title,author_id,published_year,available_copies) VALUE
rich dad poor dad',101,2001,10),(2,'Automatic Habits',102,1999,100),(3,'good habits',10
8,20),(4,'can we be stranger again',101,2000,102),(5,'Qanquer Of Destiny',102,2020,500)
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> select * from Books;
+---------+-----------+--------------------------+---------------+------------------+-
-+
| book_id | author_id | title                    | published_year | available_copies |
 |
+---------+-----------+--------------------------+---------------+------------------+-
-+
|       1 |       101 | rich dad poor dad        |          2001 |               10 |
 |
|       2 |       102 | Automatic Habits         |          1999 |              100 |
 |
|       3 |       103 | good habits              |          1978 |               20 |
 |
|       4 |       101 | can we be stranger again |          2000 |              102 |
 |
|       5 |       102 | Qanquer Of Destiny       |          2020 |              500 |
 |
+---------+-----------+--------------------------+---------------+------------------+-
-+
5 rows in set (0.00 sec)
```

10.Insert 3 members into the Members table.

```
mysql> insert into Members(member_id,name,email,phone) values(1001,'kailash','kailash@g
com',5696354885),(1002,'Vikas','Vikas@gmail.com',892456738),(1003,'Prakash','Prakash@gm
om',9787393090);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> select * from Members;
+-----------+---------+-------------------+------------+
| member_id | name    | email             | phone      |
+-----------+---------+-------------------+------------+
|      1001 | kailash | kailash@gmail.com | 5696354885 |
|      1002 | Vikas   | Vikas@gmail.com   | 892456738  |
|      1003 | Prakash | Prakash@gmail.com | 9787393090 |
+-----------+---------+-------------------+------------+
3 rows in set (0.00 sec)
```

11. Insert 4 borrow records into the Borrow_Records table.

```
mysql> select * from Borrow_Records;
+-----------+-----------+---------+-------------+-------------+
| record_id | member_id | book_id | borrow_date | return_date |
+-----------+-----------+---------+-------------+-------------+
|      5000 |      1001 |       1 | 2026-02-12  | 2026-03-12  |
|      5001 |      1002 |       2 | 2026-03-12  | 2026-04-12  |
|      5003 |      1003 |       3 | 2026-08-12  | 2026-10-12  |
|      5004 |      1001 |       1 | 2026-12-12  | 2027-05-12  |
+-----------+-----------+---------+-------------+-------------+
4 rows in set (0.00 sec)
```

12. Write an SQL query to select all books where published_year is after 2015.

```
mysql> select * from Books where published_year >= 2015;
+---------+-----------+--------------------+----------------+------------------+-----------+
| book_id | author_id | title              | published_year | available_copies | genre     |
+---------+-----------+--------------------+----------------+------------------+-----------+
|       5 |       102 | Qanquer Of Destiny |           2020 |              500 | Gen-Alpha |
+---------+-----------+--------------------+----------------+------------------+-----------+
1 row in set (0.00 sec)
```

14. Write an SQL query to find all members who have borrowed the book with title 'Database Systems'.

```
mysql> select * from Books where title ='rich dad poor dad';
+---------+-----------+------------------+----------------+------------------+-------+
| book_id | author_id | title            | published_year | available_copies | genre |
+---------+-----------+------------------+----------------+------------------+-------+
|       1 |       101 | rich dad poor dad |           2001 |               10 | Gen-Z |
+---------+-----------+------------------+----------------+------------------+-------+
1 row in set (0.01 sec)
```

15. Update the available_copies column of a specific book (choose any book) by reducing it by 1 after it is borrowed.

```
mysql> UPDATE Books
    -> SET available_copies = available_copies - 1
    -> WHERE available_copies = 500;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from Books;
+---------+-----------+--------------------------+----------------+------------------+-----------+
| book_id | author_id | title                    | published_year | available_copies | genre     |
+---------+-----------+--------------------------+----------------+------------------+-----------+
|       1 |       101 | rich dad poor dad        |           2001 |               10 | Gen-Z     |
|       2 |       102 | Automatic Habits         |           1999 |              100 | Gen-Z     |
|       3 |       103 | good habits              |           1978 |               20 | Gen-X     |
|       4 |       101 | can we be stranger again |           2000 |              102 | Gen-X     |
|       5 |       102 | Qanquer Of Destiny       |           2020 |              499 | Gen-Alpha |
+---------+-----------+--------------------------+----------------+------------------+-----------+
5 rows in set (0.00 sec)
```

16 Delete a record from Members where member_id = 3.

```
mysql> delete from Members where member_id=1003;
Query OK, 1 row affected (0.00 sec)

mysql> select * from Borrow_Records;
+-----------+-----------+---------+-------------+-------------+
| record_id | member_id | book_id | borrow_date | return_date |
+-----------+-----------+---------+-------------+-------------+
|      5000 |      1001 |       1 | 2026-02-12  | 2026-03-12  |
|      5001 |      1002 |       2 | 2026-03-12  | 2026-04-12  |
|      5003 |      NULL |       3 | 2026-08-12  | 2026-10-12  |
|      5004 |      1001 |       1 | 2026-12-12  | 2027-05-12  |
+-----------+-----------+---------+-------------+-------------+
4 rows in set (0.00 sec)

mysql>
```

17. Update a Book name record from Book table with id = 1.

```
mysql> update Books set title ='comics' where book_id=3;
Query OK, 1 row affected (0.03 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from Books where book_id=3;
+---------+-----------+--------+----------------+------------------+-------+
| book_id | author_id | title  | published_year | available_copies | genre |
+---------+-----------+--------+----------------+------------------+-------+
|       3 |       103 | comics |           1978 |               20 | Gen-X |
+---------+-----------+--------+----------------+------------------+-------+
1 row in set (0.00 sec)
```

18. Write an SQL query to list all books along with their authors' names.

```
mysql> select Books.title , Authors.name from Books join Authors on Books.Author_id = Authors.Author_id;
+--------------------------+--------+
| title                    | name   |
+--------------------------+--------+
| rich dad poor dad        | tejas  |
| can we be stranger again | tejas  |
| Automatic Habits         | khushi |
| Qanquer Of Destiny       | khushi |
| comics                   | rohan  |
+--------------------------+--------+
5 rows in set (0.01 sec)
```

19. Write an SQL query to delete all books from the Books table where the published_year is before 2000.

```
mysql> delete from Books where published_year < 1999;
Query OK, 1 row affected (0.02 sec)

mysql> select * from Books;
+---------+-----------+--------------------------+----------------+------------------+-----------+
| book_id | author_id | title                    | published_year | available_copies | genre     |
+---------+-----------+--------------------------+----------------+------------------+-----------+
|       1 |       101 | rich dad poor dad        |           2001 |               10 | Gen-Z     |
|       2 |       102 | Automatic Habits         |           1999 |              100 | Gen-Z     |
|       4 |       101 | can we be stranger again |           2000 |              102 | Gen-X     |
|       5 |       102 | Qanquer Of Destiny       |           2020 |              499 | Gen-Alpha |
+---------+-----------+--------------------------+----------------+------------------+-----------+
4 rows in set (0.00 sec)
```

20 . Write an SQL query to **find all books that are never borrowed (i.e., no records in `Borrow_Records`).**

```
mysql> select b.* from Books b left join Borrow_Records br on b.book_id = br.book_id where br.book_id is null;
+---------+-----------+--------------------------+----------------+------------------+-----------+
| book_id | author_id | title                    | published_year | available_copies | genre     |
+---------+-----------+--------------------------+----------------+------------------+-----------+
|       4 |       101 | can we be stranger again |           2000 |              102 | Gen-X     |
|       5 |       102 | Qanquer Of Destiny       |           2020 |              499 | Gen-Alpha |
+---------+-----------+--------------------------+----------------+------------------+-----------+
2 rows in set (0.06 sec)
```